

Structural Loading on Appendages

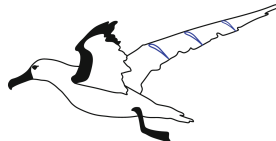
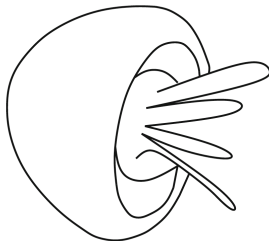
Understanding Lift, Bending, and Torsion in Natural Flyers

Motivation

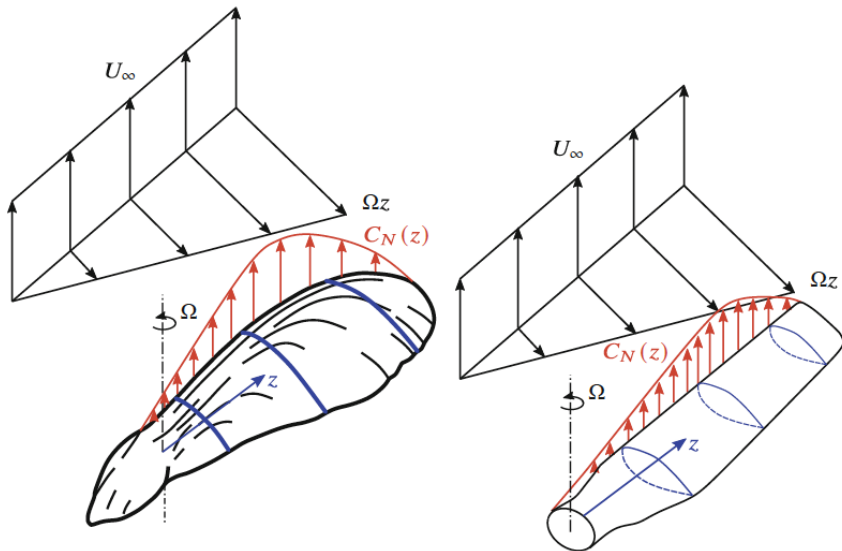
- Understand how nature handles aerodynamic forces
- Design better flying machines
- Predict and prevent structural failures

Propulsive Appendages in Nature

- Fish fins, bird wings, insect wings
- Provide thrust and control
- Need to balance strength and flexibility



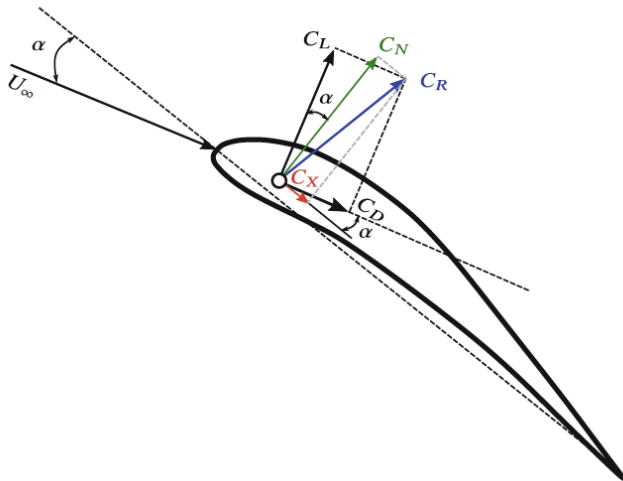
Bio-Inspired Propulsion



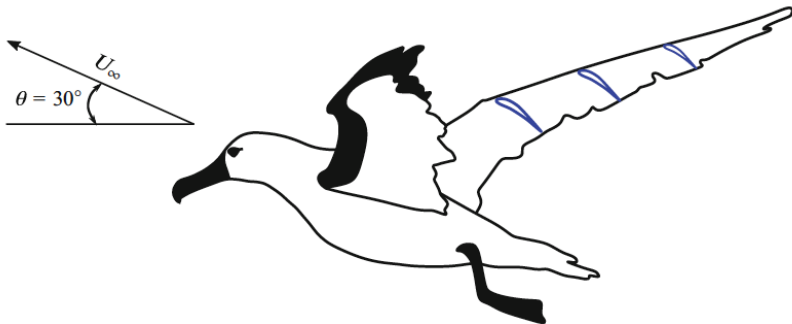
Modeling Structural Forces

- Wings split into 2D slices
- Forces: lift, drag, normal, axial
- Moments: bending (flap/edge) and torsion

Force Components on a Section



Albatross Takeoff Scenario



Takeoff Modeling Example

- Use geometry and airspeed
- Apply known coefficients (C_L , C_D)
- Compute flapwise, edgewise, and torsional moments